Comparative Prediction Analysis of Binary and Multiclass Classification Using Two Machine Learning Models

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Abstract

Goal of the project: Developing machine learning models for binary and multiclass classification on transcriptomics data and comparing them to pre-existing SVM model based on model metrics and important features.
Main results: The binary models achieved high performance metrics while the multi-class are poor in their predictive power, important features differ in all models.
Personal key learnings: Handling of multi class machine learning models, in evaluation and feature extraction specifically.
Estimated working hours: 8
Project evaluation: 3
Number of Words: 1952

Goal

The problem addressed in this machine learning task is the identification of cancer from blood-based data. This is important because it contributes to a non-invasive patient diagnosis and improvement of further treatment.
Here we focus on the use of blood-drawn transcriptomics data from the cancer cohort of study ***Best et al.*** (***2015***) to develop two binary classifiers, Random Forest and PCLA, and evaluate their performance against the previously presented SVM using AUC scores and confusion matrices. We also aim to construct multiclass versions of both models to distinguish primary cancer types and assess important features across all four models by comparing the top ten biomarkers with each other and those presented in the paper.

Data

We leveraged the Tumor-Educated Platelets (TEP) mRNA-Seq data cohort from ***Best et al.*** (***2015***), in which 6 mL EDTA blood was drawn and processed within 48 h for platelet isolation, mRNA amplification and sequencing. This resulted in 57736 transcripts over 285 samples. The cohort contained 55 healthy donors and 230 cancerous patients. The 230 cancer patients also differ in the primary cancer type: Non-small cell lung carcinoma (NSCLC) n = 59, Colorectal cancer (CRC) n = 46, Glioblastoma (GBM) n = 40, Pancreatic adenocarcinoma (PAAD) n = 34, Breast cancer (BrCa) n = 39, and Hepatobiliary cancer (HBC) n = 12.

Data Preprocessing

We downloaded the dataset from GEO, which comprised approximately 57,000 Ensembl transcript IDs as rows and patient samples labeled by the diagnosis as columns.

First, we transposed the data, so that the rows are the patients and columns represent the transcripts. Because the cancer diagnoses were inconsistently worded even when referring to the same primary tumor type, we manually standardized all labels and created new columns for the standardized primary diagnosis (cancer_type) and a binary indicator of cancer versus healthy status (condition) using metadata also available at GEO. Following the original study, we filtered out any gene with fewer than five reads in every sample, reducing the feature set to approximately 16,000 transcripts.

To ensure class balance, we downsampled each group to match the smallest class size. For the binary model, this meant reducing the tumor cohort to 55 samples, equal to the healthy group. We also evaluated an unbalanced (230 vs 55) approach because the cancer class includes multiple subtypes and disease stages we could hardly stratify easily. However, in the end, it yielded no meaningful performance gains compared to the balanced dataset, so we just kept the downsampled data approach for the binary classification. For the multiclass model, we similarly downsampled each cancer subtype to 12 samples, matching the size of the HBC cohort.

Train/Test Split

After pre-processing, both data for binary and multiclass predictions were split in train and test data, equivalent to the train and validation data sets in the original paper. We used sci-kit-learn's train_test_split with an 80-20 train-test data proportion. Later, we also scaled the raw read counts using sci-kit-learn's StandardScaler to center each feature at zero mean and unit variance, a crucial preprocessing step for read-count expression data since the raw expression counts vary greatly.

Method

After data preprocessing, we train two machine learning models, Random Forests and Principal Component Linear Regression Analysis, on the two classification datasets, resulting in four different models. Model training and downstream evaluation were implemented using scikit-learn (***Pedregosa et al., 2018***).

Random Forests

Random Forests is a method that constructs multiple decision trees from the data and combines their results to predict the output (***Chen et al., 2013***). For the model implementation, RandomForestClassifier from scikit learn was used. Model parameters were set to default. We refrained from hyperparameter optimization, as the default model already yielded good results and a larger hyperparameter search space would make the training process more time-consuming.

For each classification, a separate model with the same initial settings is trained and evaluated iteratively. After training, we identify the top ten most important features for each model variant using the feature_importances_ attribute, which quantifies the contribution of each feature to the model's predictive performance based on the total decrease in node impurity (***Chen et al., 2013***). For the multiclass feature extraction, the values signify the feature importance of features across all classes.

Principal Component Linear Regression Analysis

PCLR chains two ML techniques: first, PCA reduces the original high-dimensional data down to the top N principal components, and then a standard logistic regression is fit on these component scores. This is perfect for any data with many correlated features, since PCA gives uncorrelated axes before an LR is trained. We optimized N via a randomized search using sci-kit-learn's RandomizedSearchCV and implemented PCA and LR in a Pipeline ***Pedregosa et al.*** (***2018***). For binary classification, we used the "liblinear" solver—an efficient method for small datasets, while for multiclass tasks, we switched to "lbfgs" since "liblinear" cannot handle the multinomial loss. In addition,

we evaluated the feature's importance by mapping the logistic regression coefficients of our principal components back to the original genes. Finally, we sorted the resulting weights per gene by absolute values. For multi-class feature importance, these values were averaged over all classes.

Evaluation

Performance evaluation is an important step in assessing our model's reliability. For binary models, we calculated the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) score for both train (validation) and test data and accuracy on both train and test data. Additionally, we plotted ROC-curves and confusion matrices for both RF and PCLA models.
In case of multi-class prediction, we calculate the accuracy score and plot confusion matrices.

Results & Discussion

Model Metrics

First, we compare model evaluation metrics for four of our models and subsequently compare to the SVM results described in the original paper. The metrics evaluation are shown in ***Table A1*** and ***Figure A1–Figure A3***.
Both RF and PCLA models achieve an AUC_{train} score of 1, indicating perfect class separation in training. The AUC_{test} for the PCLA model is slightly higher than that of the RF model. Still, yet both values exceed 0.95. Additionally, their accuracy scores also indicate strong predictive value. Together, AUC_{test} and accuracy scores signify both models with high predictive power when classifying healthy platelets vs. TEPs. Confusion matrices reveal that on the test set, confusion matrices differ by one missclassied healthy sample in the RF model.
In comparison to the AUC scores from the binary classification in the paper, both RF and PCLA AUC_{train} scores are higher than that of the paper, each achieveing a perfect score of 1. In terms for AUC_{test} (referred to as $AUC_{validation}$ in the paper), the PCLA outperforms the SVM model, while the RF model performs slightly worse.

However, for multi-class prediction, the performance drops sharply. While all training samples were classified correctly, as shown by perfect confusion matrices and accuracy scores, test set predictions resulted in low accuracy. Between the two, RF performs slightly better than PCLA on test accuracy, yet both models underperform. From the test data confusion matrices, the platelets from the healthy donor could be classiefied most accurately. Furthermore, the multi-class RF model could correctly predict HBC platelets two out of three times. while as for the PCLA model also predicts two out of three healthy platelets and classifies all PAAD platelets correctly. This big gap between accuracy in training and test data suggests both models are overfitting. Compared to RF and PCLA models, the SVM multi-class model in the paper shows better predictive performance on the test (validation) data. Although not perfect, it still holds greater predictive power than the RF and PCLA models.

Important Features

In order to compare elucidated features in both our and the SVM model used in the paper, we use the supplementary tables S2 for binary and S5 for multi-class prediction. Those contain differentially expressed genes sorted by their p-values and include logFold values. Additionally, we refer to table S4 to check features (transcripts) used in the SVM model. The results from our top 10 feature extraction can be seen in ***Table A3–Table A6***.
First, the features important to the RF and PCLA classifications differ entirely as there are no overlapping transcripts. Additionally, in the top ten features extracted from the binary PCLA model, there are five non-coding RNAs, different to the features from the RF model where all transcripts correspond to a protein-coding RNA.
When compared to the differentially expressed genes in the healthy platelets vs. TEP both features from RF and PCLA binary models could be found. Although not in the top 10 differentially

expressed genes, those genes are still found to be distinctive between healthy platelets and TEPs. Interestingly enough, not all of those features to be predictively important are used in the gene list of the binary SVM model.

For both multi-class models, no features overlap as well. Both models contain non-coding RNA in their top 10 features - RF two and PCLA three.
When comparing the differentially expressed genes, we compare one table each for RF and PCLA models, which contain averaged feature importances, to tables that contain feature importance for healthy vs. each cancer type.
We see that six transcripts from the RF feature list are present across cancer types' differentially expressed genes, except for GBM, where none of the transcripts from both tables could be found. However, none of the PCLA transcripts appeared in any of the gene tables. Overall, fewer features are found in the differentially expressed genes table compared to the binary prediction, and similar to the binary case, not all of them reappear in the gene list of the SVM model.

Main Insights

Overall, the binary classification task produced better results for both RF and PCLA, as well as for the SVM model from the original paper. This may be caused by two factors: first, the distinguishing factors of healthy platelets and TEPs in general might be more pronounced than those of singular cancer types. This is also reflected in both binary and multi-class confusion matrices, where the healthy or "none" class is the one to be predicted more accurately among others. Second, the poor performance of models examined here could be attributed to a very low number of samples. The multi-class models had about 10 'training' samples per class, whereas for test prediction there were only two to three available. Such smaller number of samples or limited data can cause the prediction influenced by the presence of noise, resulting in overfitting and poor predictive performance on unseen data. Moreover, SVM is better suited to work with high-dimensional data - in our case about 16000 transcript features - and small sample sizes (***Huang et al., 2018***).

Regarding the important features and differentially expressed genes (DEGs) in the case of binary models, all features were found in the DEGs table. These, however, were different between RF and PCLA models. This likely reflects the inherently different approach of the models when it comes to training and feature extraction. GIven the high number of features in the data, one model may identify one combination of biological signals whereas the other relies on a different combination. As for biological interpretation, evidence of some of the genes in cancer biology could be found. For example, the first ranking transcript of the RF list ENSG00000198851 (CD3e), a subunit in the T-cell receptor, is shown to be downregulated in some tumors (***Prado-Garcia et al., 2007***). In the list of PCLA transcripts, some correspond to cancer. For example, ENSG00000183763 (TRAIP) gene is involved in damage response and cell growth and also was found to be related to cancer progression (***Guo et al., 2020***). However, none of these transcripts have known direct involvement with TEPs.

For the multi-class models, which performed poorly, some important features—especially in the RF model—also appeared in the DEG lists. This may be partially attributed to the fact that we used averaged feature tables for both, which could exclude some relevant features. But when keeping in mind that both models performed poor on the classification task, any type of features extracted from RF and PCLA may not hold any reliability and any biological interpretation of them should be approached with caution.

Contributions

1. Part of Data processing and Random Forest model were implemented by Ksenia Krasikova.
2. All model metrics and feature importances were implemented by Payal Priyadarshini
3. Part of Data processing and PCLA model were implemented by Philipp Harlos

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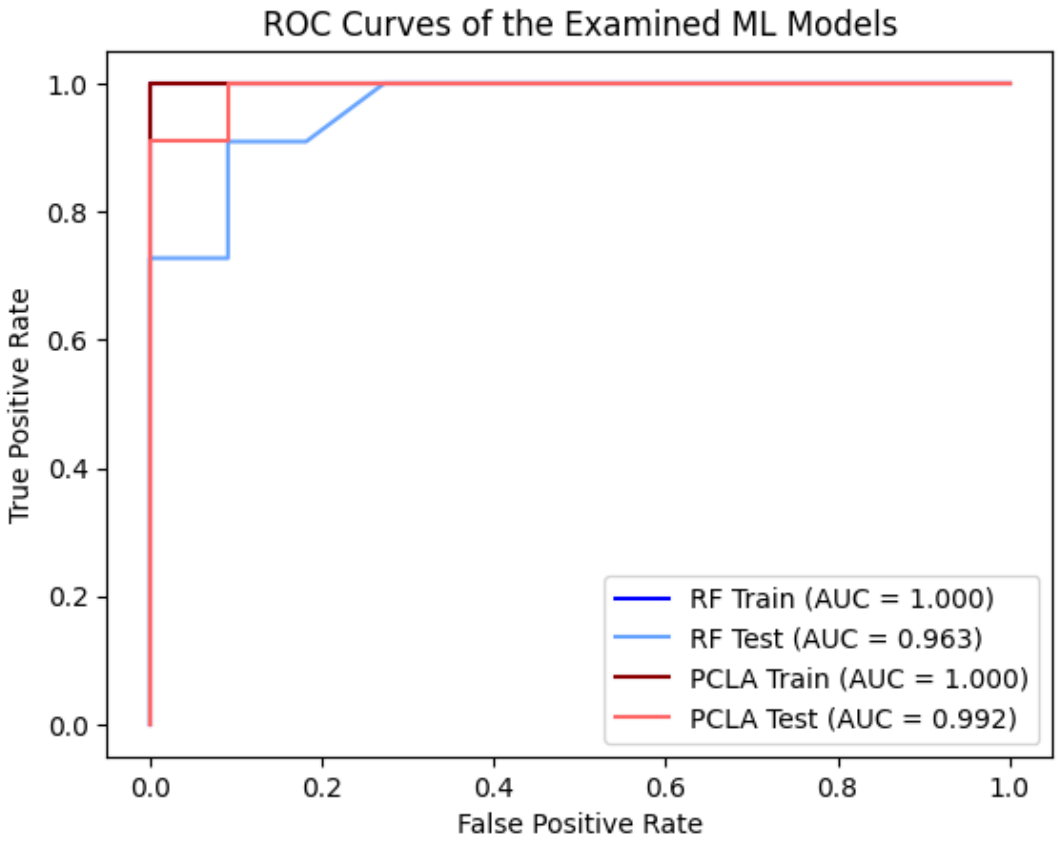
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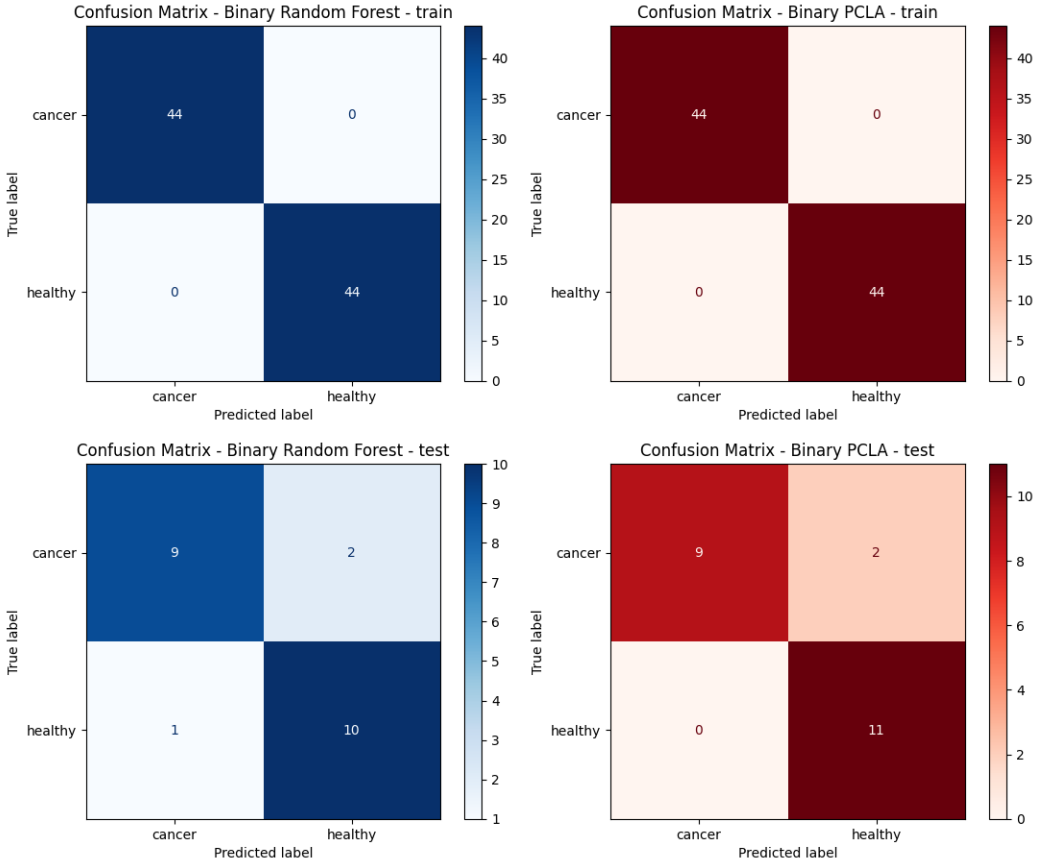
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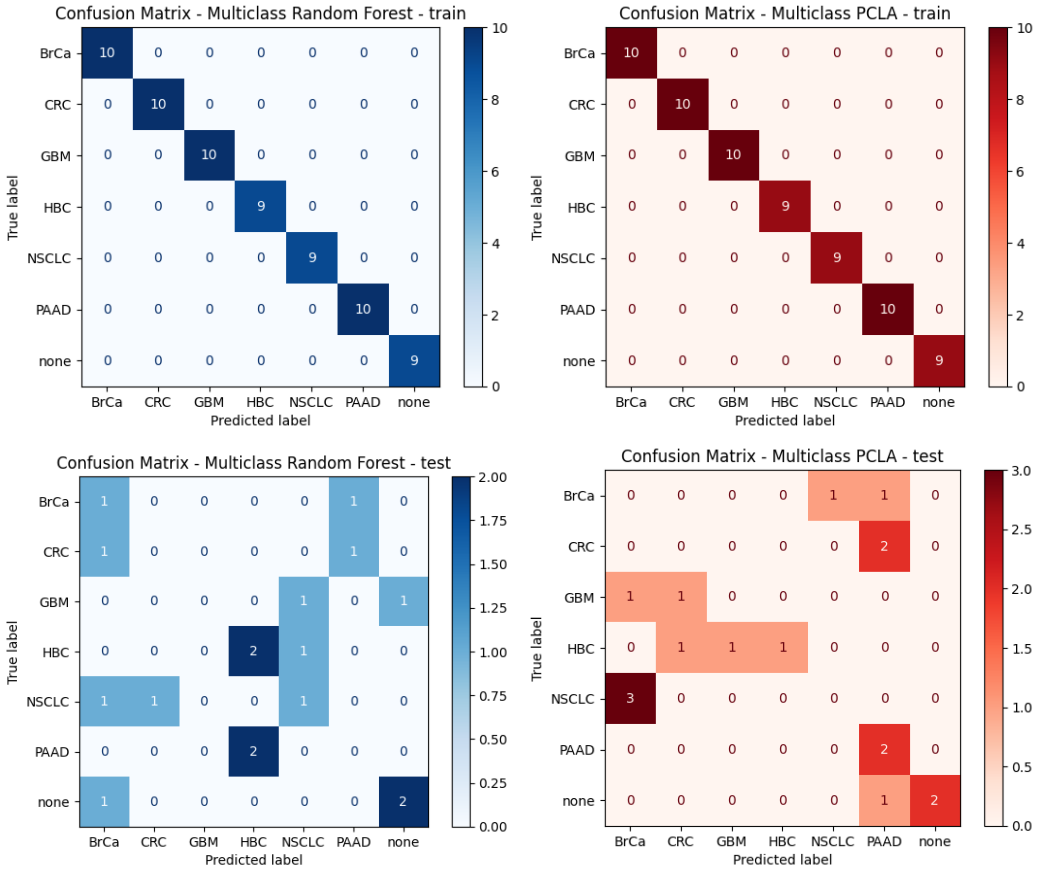
Supplementary Figures



Appendix 0—figure A1. ROC curves for our two models—Random Forest and PCLA. Blue lines represent RF model performance on train and test data whre the red lines show PCLA preformance. The legend in each plot shows its corresponding AUC score.



Appendix 0—figure A2. Confusion matrices for binary Random Forest and PCLA models (top: train dataset, bottom: test dataset).



Appendix 0—figure A3. Confusion matrices for multi-class Random Forest and PCLA models (top: train dataset, bottom: test dataset).

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Supplementary Tables

Appendix 0—table A1. Performance of Random Forest Models with Different Data Conditions

Model	AUC (Train)	AUC (Test)	Accuracy (Train)	Accuracy (Test)
Random Forest (Binary)	1.000	0.963	1.000	0.864
Random Forest (Multi)	—	—	1.000	0.353
PCLA (Binary)	1.000	0.992	1.000	0.909
PCLA (Multi)	—	—	1.000	0.294

Appendix 0—table A2. Classification Results per Subtype for binary Random Forest and PCLA Models

Subtype	RF (Correct/Total)	RF (%)	PCLA (Correct/Total)	PCLA (%)
BrCa	2/2	100%	2/2	100%
GBM	3/5	60%	3/5	60%
NSCLC	4/4	100%	4/4	100%
None	10/11	91%	11/11	100%

Appendix 0—table A3. Top 10 features for Random Forest – Binary

Rank	Ensembl ID	Importance
1	ENSG00000198851	0.0133
2	ENSG00000122224	0.0104
3	ENSG00000184922	0.0094
4	ENSG00000133422	0.0085
5	ENSG00000152601	0.0085
6	ENSG00000168685	0.0081
7	ENSG00000074966	0.0079
8	ENSG00000159461	0.0074
9	ENSG00000174917	0.0071
10	ENSG00000142192	0.0064

Appendix 0—table A4. Top 10 features for Random Forest – Multiclass

Rank	Ensembl ID	Importance
1	ENSG00000102898	0.0048
2	ENSG00000153904	0.0038
3	ENSG00000228651	0.0037
4	ENSG00000166037	0.0037
5	ENSG00000010327	0.0032
6	ENSG00000165502	0.0031
7	ENSG00000255002	0.0030
8	ENSG00000110651	0.0030
9	ENSG00000145734	0.0030
10	ENSG00000166432	0.0029

Appendix 0—table A5. Top 10 features for PCLA – Binary

Rank	Ensembl ID	Weight
1	ENSG00000269161	0.0392
2	ENSG00000266935	0.0381
3	ENSG00000254026	0.0375
4	ENSG00000197140	0.0371
5	ENSG00000266371	0.0360
6	ENSG00000272865	0.0356
7	ENSG00000183763	0.0353
8	ENSG00000088305	0.0351
9	ENSG00000103154	0.0349
10	ENSG00000131914	0.0348

Appendix 0—table A6. Top 10 features for PCLA – Multiclass

Rank	Ensembl ID	Weight
1	ENSG00000220785	0.011 094
2	ENSG00000135773	0.011 094
3	ENSG00000260475	0.011 094
4	ENSG00000136531	0.011 094
5	ENSG00000022355	0.011 094
6	ENSG00000126785	0.011 094
7	ENSG00000138395	0.011 094
8	ENSG00000258689	0.011 094
9	ENSG00000231666	0.011 075
10	ENSG00000136155	0.011 075